Image Analysis Report

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Executive Summary

The inspection reveals severe localized corrosion and coating failure concentrated on a weld or joint, consistent with a high severity rating. The presence of thick oxide scale and significant surface degradation suggests advanced metal loss and potential pitting that requires immediate NDT evaluation.

Key Findings

#	Finding
1	Heavy oxide scale buildup indicating active and advanced localized corrosion.
2	Failure of the protective coating system with visible delamination and blistering around the primary corrosion site.
3	Preferential corrosion occurring along a horizontal feature, likely a weld seam or plate junction.
4	Evidence of rust 'bleeding' indicating consistent exposure to moisture and atmospheric contaminants.
5	Appearance of dark brown/black oxidation suggesting deep, localized pitting in the central area.
6	Remaining wall thickness cannot be determined from the image alone.
7	The depth of the pits cannot be accurately measured without physical tools or NDT.

Recommendations

#	Recommendation
1	Perform Ultrasonic Testing (UT) to determine the percentage of wall loss and remaining structural integrity.
2	Conduct Pit Depth Measurements using a calibrated pit gauge after cleaning the surface.
3	Abrasive blast the area to Sa 2.5 (near-white metal) to remove all corrosion products and prepare for repair.
4	If wall loss exceeds the corrosion allowance, perform a weld overlay or apply a structural composite wrap per ASME PCC-2
5	Carry out Magnetic Particle Inspection (MPI) to check the integrity of the weld seam for stress-corrosion cracking.
6	Apply a high-performance coating system, including a surface-tolerant epoxy primer and a UV-resistant topcoat.
7	Increase the inspection frequency for this specific asset to a 12-month interval until coating integrity is stabilized.